

Lab Exercise 2.3: Ping Sweeping and Port Scanning Attacks and Defense**Learning Outcomes**

Upon completion of this laboratory exercise, you should be able to:

- Conduct and defend against ping sweeping and port scanning attacks

Required Hardware

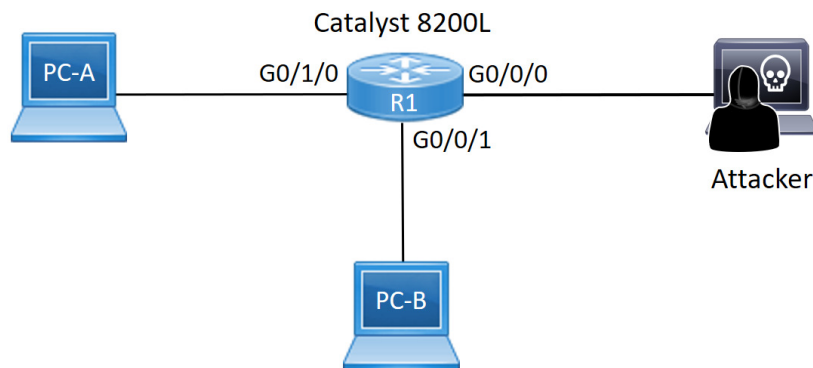
- 1 x Rack of Cisco networking devices
- 1 x Box of cables containing
 - USB-to-RJ45 console cables
 - Ethernet cables
- 2 x Laptops

Required Software

- Tera Term <https://teratermproject.github.io/index-en.html>
- Kali Linux Live Boot on USB drive

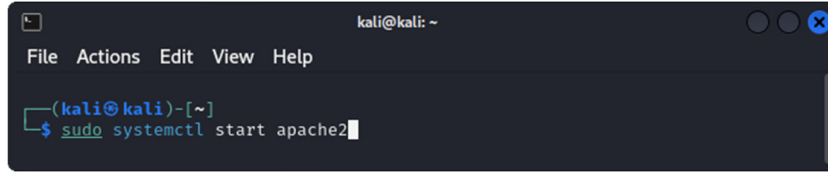
PART 1: Conducting Ping Sweeping and Port Scanning using Nmap

1.1 Setup the network topology as shown.



- 1.2 Boot up PC-A in Windows to simulate user inside enterprise LAN; boot up PC-B in Kali Linux to be Web server in the DMZ; and boot up Attacker's PC in Kali Linux to simulate potential attacker from the Internet.
- 1.3 Configure suitable IP addresses on PC-A, PC-B, Attacker's PC and router R1's interfaces so that they are able to ping one another successfully.

- 1.4 On PC-B, start up a terminal and run the apache web server with default web page as shown:



```
kali@kali: ~
File Actions Edit View Help

(kali@kali)~$ sudo systemctl start apache2
```

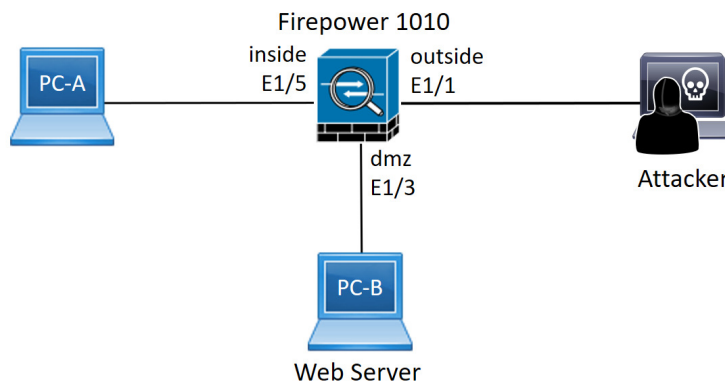
- 1.5 When ready, scan the subnets of PC-A and PC-B from Attacker's PC using nmap with different options as discussed. Observe the results.

PART 2: Defending Ping Sweeping and Port Scanning using Router with ACL

- 2.1 On Attacker's PC, start up a terminal and also run the apache web server with default web page to simulate web server in the Internet.
- 2.2 Apply suitable ACL on R1 G0/0/0 to allow Attacker to visit PC-B web server hosted at DMZ, but not PC-A inside enterprise LAN.
- 2.3 When ready, scan the subnets of PC-A and PC-B from Attacker's PC again. Observe the results and compare with that in Step 1.5.
- 2.4 Do you think router with ACL is effective against ping sweeping and port scanning attack?

PART 3: Defending Ping Sweeping and Port Scanning using Firewall

- 3.1 Replace the router with a firewall in the network topology as shown.



- 3.2 For simplicity, configure similar IP addresses on the firewall as the router.
- 3.3 Similarly, apply suitable ACL on the firewall to allow Attacker to visit PC-B web server hosted at DMZ, but not PC-A inside enterprise LAN.

- 3.4 When ready, scan the subnets of PC-A and PC-B from Attacker's PC again. Observe the results and compare with that in Step 1.5 and 2.3.
- 3.5 Do you think firewall is better than router with ACL in defending against ping sweeping and port scanning attack?
- 3.6 Is there any way to prevent scanning the Web server at the DMZ and yet allowing user from the Internet to visit the website?